Lesson 2.4

Algebraic Reasoning & Properties

Every step needs a reason

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Every step needs a reason

In geometry we don't just get the answer — we **justify** every step with a **reason**: a definition, a property, a postulate or a theorem.

Solve and justify

$x - 4 = 10$ Given

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$x = 14$	Addition Property of Equality

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The reason names **what you did to both sides**.

Properties of equality

Addition / Subtraction: add or subtract the same number on both sides.
Multiplication / Division: multiply or divide both sides by the same number (not 0).

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Substitution: if $a = b$, you can replace a with b.
Distributive: $a(b + c) = ab + ac$.

Reflexive, symmetric, transitive

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Symmetric: if $a = b$, then $b = a$.

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Symmetric: if $a = b$, then $b = a$.

Transitive: if $a = b$ and $b = c$, then $a = c$.

A longer example

$3(x + 2) = 21$ Given

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$3(x+2)=21$	Given
$3x+6=21$	Distributive Property

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$3(x+2)=21$	Given
$3x+6=21$	Distributive Property
$3x=15$	Subtraction Property of Equality

A longer example

$3(x+2)=21$	Given
$3x+6=21$	Distributive Property
$3x=15$	Subtraction Property of Equality
$x=5$	Division Property of Equality

Your turn! Pause the video and try it.

Name the property.

(a) If $AB = CD$, then $CD = AB$.

(b) If $m\angle 1 = m\angle 2$ and $m\angle 2 = m\angle 3$, then $m\angle 1 = m\angle 3$.

Your turn! Pause the video and try it.

Name the property.
(a) If $AB = CD$, then $CD = AB$. (b) If $m\angle 1 = m\angle 2$ and $m\angle 2 = m\angle 3$, then $m\angle 1 = m\angle 3$.

(a) Symmetric — the equation is flipped. (b) Transitive — a chain.

Remember

- Every step needs a **reason**.
- Did something to both sides? Name that property of equality.
- **Reflexive** $a = a$, **symmetric** flip, **transitive** chain.

Now practice at your level:

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